PARKHI.ai - Technical Summary

PARKHI.ai is a fraud-detection and transaction analysis platform designed to help users upload raw financial data, process it through an automated machine learning pipeline, and review the results through an interactive dashboard. The project is built to handle noisy, real-world CSV data and convert it into structured, explainable fraud insights that are easy to review and act on.

The system is organized into two primary parts: a Next.js frontend and a Python-based fraud pipeline. The frontend, located in client/, serves as the main user interface. It includes a landing page, dashboard pages, fraud report views, KPI summaries, EDA visualizations, and feeds. The dashboard is designed to make the analysis workflow simple for end users, so they can upload a transaction file, wait for processing, and then inspect the output in a clean visual format.

The fraud pipeline, located in mlpy/, is the core analytical engine of the project. It is built with FastAPI and runs a complete processing workflow on uploaded CSV files. The pipeline first cleans and standardizes transaction data by handling inconsistencies, missing values, duplicate records, and formatting issues. After cleaning, it detects suspicious patterns and engineers features that help identify fraudulent behavior. These features may include transaction amount deviation, unusual transaction frequency, odd-hour activity, device novelty, and location mismatch signals. Once the features are prepared, the model scores each transaction for fraud risk and applies thresholding to determine which records should be flagged. The pipeline also generates metrics, EDA charts, and a fraud report that summarizes the analysis in a readable format.

A major strength of PARKHI.ai is its output structure. Instead of returning only raw predictions, the system produces multiple artifacts that support review and interpretation. These include a fraud_report.json file, EDA chart outputs, cleaned transaction files, scored transaction summaries, and KPI data. The frontend consumes these outputs through internal API routes and presents them in dedicated pages. This makes the project useful not only as a detection engine, but also as a reporting and decision-support tool.

The overall workflow is straightforward. A user uploads a CSV file in the frontend, the request is forwarded to the Python pipeline, the dataset is cleaned and analyzed, fraud scores are generated, and the results are displayed back in the dashboard. This separation between interface and pipeline keeps the application modular, easier to maintain, and easier to extend in the future.

PARKHI.ai uses a modern technology stack suited for full-stack data applications. The frontend is built with Next.js, React, TypeScript, and Tailwind CSS. The analysis pipeline is built with Python, FastAPI, pandas, NumPy, scikit-learn, LightGBM, and Matplotlib. Together, these tools enable both strong analytical processing and a polished user experience.

In summary, PARKHI.ai is a fraud analytics platform that transforms raw transaction data into explainable risk intelligence. It combines data cleaning, feature engineering, fraud scoring, EDA generation, and dashboard visualization into one streamlined workflow, making it a practical solution for reviewing suspicious financial activity.